ASSIGNMENT

Gasification & Reforming Processes

Group Formation

Each topic will be assigned to **6 groups**. Each group must consist of a **maximum of 5 students**. Each group will select one of the following topics:

1. Steam Methane Reforming
2. Fixed-Bed Gasifier
3. Fluidised-Bed Gasifier
4. Plasma Gasifier
5. Entrained-Flow Gasifier

Assignment Requirements

Each group must refer to relevant **peer-reviewed journal papers** for their selected topic. Groups working on the **same topic must select different journal papers** and use **different operating conditions**. No two groups should use the same journal paper or identical operating conditions.

The selected journal paper and its operating conditions must be clearly stated in the report. These conditions should be used as the basis for the simulation, and the simulation results must be **validated against the corresponding journal results**.

Required Results

- Feedstock/feed composition and major species
- Species distribution graphs
- Species contour plots inside the reactor
- Reactor temperature/temperature contours, where applicable
- Comparison of simulation results with the selected journal results
- Conclusion summarising the major findings

Submission Prepare a final **report/document** containing the required results, graphs, contours, validation, discussion, and conclusion. Submit the final report through the **Google Drive link provided in the Excel (XL) sheet.**

The submitted report must be named using the **Batch Number** (e.g., `Batch_01.pdf`). In the same Google Drive folder, each group must also upload the relevant **simulation/source files**, including the **Python or MATLAB code**

CLASS PRESENTATION

Each group must present their work in class starting from mid-October. The presentation must include all the above-mentioned results, graphs, species contours, validation, and conclusions.

Maximum Marks	40
Group Evaluation	All group members will receive the same marks
Deadline	MId of October 2026
